

Sharp FRS and singularity thresholds for off-diagonal Gram maps

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Abstract

The off-diagonal Gram map for h vectors in an M -dimensional quadratic space is flat with reduced fibers of rational singularities (FRS) in characteristic zero if and only if $M > \max_{2 \leq r \leq h} (2h - r + 1 - 2h/r)$. Equivalently, this gives the exact FRS convolution threshold of $x \mapsto (x_i x_j)_{i < j}$. For its zero-fiber ideal $I_{M,h}$, we also prove

$$\text{lct}(I_{M,h}) = \min \left\{ \binom{h}{2}, \min_{0 \leq k \leq h-2} \left(\binom{k+1}{2} + \frac{M(h-k)}{2} \right) \right\}.$$

In the strict range, all truncated coefficient maps have geometrically integral complete-intersection fibers and are faithfully flat over $\mathbb{Z}[1/2]$, uniformly over graphs on h vertices. The proof uses a sharp weighted estimate for hollow symmetric matrices over finite local principal ideal rings, paired with isotropic lower bounds. Within the established point-counting approach to rational singularities, this estimate also gives an explicit modulus of continuity for Gram-map densities over every local field of odd residue characteristic.

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1 Introduction

Pairwise orthogonality imposes a structured family of quadratic equations on an ordered collection of vectors. These equations define orthogonal frame schemes; choosing only selected pairs gives the Lovász–Saks–Schrijver ideals associated with graphs [13, 6]. Even when the full frame scheme is a normal complete intersection, its singularities need not be rational. The problem is to determine precisely how large the ambient quadratic space must be to obtain rational singularities, and to describe the singularities below that threshold.

We determine the exact FRS threshold for every number of vectors using quadratic Fourier sums over finite local principal ideal rings. The estimates include all Smith valuation patterns and are uniform in the truncation length and the prescribed Gram entries. Besides the rationality classification, the argument determines the log canonical threshold of the Gram ideal even when the zero scheme is reducible or nonreduced. It also gives integral coefficient fibers for arbitrary prescribed inner products and for arbitrary subgraphs in a sharp range uniform over graphs.

1.1 Notation and main results

Fix integers $h \geq 2$ and $M \geq 1$, and put $e = \binom{h}{2}$ and $N = Mh$. For an $M \times h$ matrix $V = (v_{ai})$, let

$$\Phi_{M,h}(V) = \left(\sum_{a=1}^M v_{ai} v_{aj} \right)_{i < j}.$$

Write $I_{M,h}$ for the ideal generated by these e polynomials and $X_{M,h} = \text{Spec } \mathbb{C}[V]/I_{M,h}$. Thus the scheme structure, and not only its reduced support, is part of the notation. We use the standard nondegenerate symmetric bilinear form. Over an algebraically closed field of characteristic different from two, every nondegenerate symmetric form is equivalent to this one. Singularity invariants are taken over $\mathbb{C}$; the corresponding characteristic-zero statements are geometric statements after extension to an algebraic closure. The scheme $(X)_{L-1}$ denotes the jet scheme representing maps $\text{Spec } A[t]/(t^L) \rightarrow X$.

For an ideal I on a smooth variety Y , we use the convention

$$\text{lct}_Y(I) = \inf_E \frac{A_Y(E)}{\text{ord}_E(I)},$$

where $A_Y(E)$ is the log discrepancy and the infimum is over prime divisors over Y with positive order on I . No reducedness convention is imposed on the ideal. We write $\text{lct}(I)$ when the ambient affine space is understood. Define

$$\delta_h(r) = \frac{r(2h-r+1)}{2} - h, \quad 2 \leq r \leq h, \quad (1.1)$$

$$F(h) = \max_{2 \leq r \leq h} \frac{2\delta_h(r)}{r} = \max_{2 \leq r \leq h} \left(2h - r + 1 - \frac{2h}{r} \right), \quad (1.2)$$

$$D_{\text{rat}}(h) = \lfloor F(h) \rfloor + 1. \quad (1.3)$$

Following Aizenbud–Avni [1], a morphism between smooth characteristic-zero varieties is *FRS* if it is flat and every geometric fiber is reduced and has rational singularities.

Theorem 1.1. *The following are equivalent:*

1. $M > F(h)$, or equivalently $M \geq D_{\text{rat}}(h)$;
2. the off-diagonal Gram map $\Phi_{M,h}$ is *FRS*;
3. its zero fiber $X_{M,h}$ has rational singularities.

In this range every fiber is an integral complete intersection with rational singularities. The same sufficient condition holds for every graph-Gram map on h vertices, obtained by retaining any subset of the output coordinates. It is sharp as a condition uniform over all such graphs.

For $\phi_h : \mathbb{A}^h \rightarrow \mathbb{A}^e$, $\phi_h(x) = (x_i x_j)_{i < j}$, the sum $\phi_h(x^{(1)}) + \cdots + \phi_h(x^{(M)})$ is precisely $\Phi_{M,h}$. Thus Theorem 1.1 determines the least FRS convolution power of ϕ_h : it is $D_{\text{rat}}(h)$. The graph statement concerns a condition valid for *all* graphs on h vertices, not a necessary threshold for each individual graph.

Theorem 1.2. *For all $M \geq 1$ and $h \geq 2$, the log canonical threshold in $\mathbb{A}_{\mathbb{C}}^N$ is*

$$\text{lct}(I_{M,h}) = \min \left\{ e, \min_{0 \leq k \leq h-2} \left(\binom{k+1}{2} + \frac{M(h-k)}{2} \right) \right\}. \quad (1.4)$$

The global threshold equals the threshold at the origin.

For $L \geq 1$, expand $V(t) = \sum_{a=0}^{L-1} V_a t^a$ modulo t^L and let $\Phi_{M,G}^{[L]}$ record all L coefficients of each inner product indexed by an edge of a simple graph G on h vertices.

Theorem 1.3. *If $M > F(h)$, then for every G and L the morphism*

$$\Phi_{M,G}^{[L]} : \mathbb{A}_{\mathbb{Z}[1/2]}^{LMh} \longrightarrow \mathbb{A}_{\mathbb{Z}[1/2]}^{L|E(G)|}$$

is faithfully flat with geometrically integral complete-intersection fibers of dimension $L(Mh - |E(G)|)$.

1.2 Relation to earlier work

The relation between finite-ring point counts and rational singularities is an established framework. Aizenbud–Avni [2] connect these counts with rational singularities and representation growth; Glazer [9] completes the criterion for local complete intersections. Cluckers–Glazer–Hendel [5] give a relative characterization uniform over the target, and Aizenbud–Avni [1] relate FRS maps to continuous densities of pushforward measures over local fields. Glazer–Hendel [10, Theorem 1.7] prove eventual FRS regularity under convolution when the image spans the target affine space. This applies to ϕ_h , since $\phi_h(0) = 0$ and $\phi_h(e_i + e_j)$ runs through the standard basis of $\mathbb{A}^e$. Our convolution identity places the Gram family in this context and gives an exact exponent for this map. The general point-counting-to-singularity principle is not new here.

The specific input is Lemma 2.4, which controls the full weighted sum over hollow Smith strata with the sharp exponential rate. Its uniformity in the ring and the conductor, together with matching isotropic lower bounds, yields the threshold classification. It also gives quantitative density estimates at every odd residue characteristic (Proposition 3.3), without discarding small odd primes. Our direct passage through Lang–Weil, constructibility, and jets records the additional universal coefficient-fiber statement over $\mathbb{Z}[1/2]$. Related uses of jet dimensions, rational singularities, and arithmetic counts occur for quiver moment maps in work of Budur [3] and Vernet [18].

The graph ideals originate in orthogonal representations studied by Lovász–Saks–Schrijver [13]; Conca–Welker [6] develop their algebraic relation to coordinate sections of determinantal varieties. For generic determinantal ideals, Johnson [11] computes multiplier ideals and log canonical thresholds, and Docampo [7, Theorem 5.6] obtains the threshold via arcs. These give the relevant determinantal context, but their generic-minor formula does not directly compute the threshold of the off-diagonal Gram ideal.

Casabella and Sammartano determine the complete-intersection and prime thresholds for orthogonal frames [4, Theorems 1–2]. In particular, their prime threshold is

$$D_{\text{prime}}(h) = \min \left\{ 2 \left\lfloor \frac{2h+1-\sqrt{8h+1}}{2} \right\rfloor + 2, 2 \left\lfloor \frac{2h+1-\sqrt{8h-7}}{2} \right\rfloor + 1 \right\}, \quad (1.5)$$

and $M \geq D_{\text{prime}}(h)$ gives a normal complete intersection. They ask whether this range already gives (F -)rational singularities [4, Question 8.1]. Theorems 1.2 and 1.1 give the sharp replacement in characteristic zero. The nonrational examples at the exceptional prime thresholds were established in [19, Theorem 1.3 and Sections 6–7] using generic transverse singularities. Section 6 gives a first-jet proof of the same obstruction. The earlier paper also studies defining ideals and depth of principal components, and proves non- F -rationality for its exceptional family in every odd characteristic. Those results are distinct from the positive classification proved here. The positive estimate closes the gap left by these counterexamples and also yields the exact ideal threshold and the coefficient-fiber theorem. The rationality criterion is proved independently of the prime-threshold theorem. We use [4] for the comparison with the normal range, including Table 1 and the normal-range corollaries. The positive argument does not use the principal-component results of [19].

For complete graphs with $h \geq 2$, the sufficient condition $M \geq 3h - 4$ follows from [17, arXiv version, Lemma 2.5 and Theorem 3.2], since $\text{pmd}(K_h) \leq 2h - 3$ and $k(K_h) = h - 1$. For sparse graphs, graph-specific bounds can be smaller than the uniform bound proved here; sharpness here means sharpness over all graphs on h vertices, including the complete graph. The present threshold has asymptotic form $2h - 2\sqrt{2h} + O(1)$. Our positive-characteristic conclusion is Theorem 1.3; no claim of strong F -regularity or F -rationality in that full range is made. The sharp positive-characteristic sufficiency question remains open.

1.3 Boundary cases and examples

The distinction between normality, log canonicity, and rationality is particularly small but nontrivial. We prove that $D_{\text{rat}}(h)$ is always $D_{\text{prime}}(h)$ or $D_{\text{prime}}(h) + 1$. In the normal range, $X_{M,h}$ is log canonical exactly when $M \geq F(h)$. Consequently the normal log-canonical but nonrational cases are exactly

$$h = 2a^2, \quad M = (2a - 1)^2, \quad a \geq 2.$$

The first is $(M, h) = (9, 8)$. In contrast, $(M, h) = (15, 12)$ is normal but not log canonical: its ideal threshold is $131/2$, whereas its codimension is 66. For $(M, h) = (3, 4)$, well outside the normal range, the ideal threshold is $11/2$.

Table 1 illustrates the distinction. Its log-canonical column is the smallest M in the normal range with log-canonical singularities, namely $\max\{D_{\text{prime}}(h), \lceil F(h) \rceil\}$.

h	Normal	Log canonical	Rational
2	2	2	2
3	3	3	3
4	4	4	4
6	7	7	7
7	8	8	8
8	9	9	10
11	14	14	14
12	15	16	16
16	22	22	22
18	25	25	26

Table 1: Selected sharp ambient-dimension thresholds for the full zero fiber.

The exact threshold is also detected by the zeroth and first jet schemes. Writing $c = \text{lct}(I_{M,h})$, we show

$$c = N - \max \left\{ \dim X_{M,h}, \frac{1}{2} \dim(X_{M,h})_1 \right\}.$$

Moreover, $\dim(X_{M,h})_{L-1} = L(N - c)$ whenever M or L is even. In the normal range the first jet alone detects rationality and log canonicity. These properties are consequences of the structure of this family, rather than general properties of jet schemes.

1.4 Proof strategy

The proof has three parts. First, a unit diagonal pivot estimates weighted counts of arbitrary symmetric matrices over finite local rings. A two-by-two unit off-diagonal pivot reduces the hollow case to that estimate; the residual diagonal constraints are scalar product equations $bc = t$. Second, Fourier inversion gives counts of prescribed Gram fibers. Lang–Weil and a separate smooth-point argument turn these counts into integral complete-intersection coefficient fibers, and constructibility passes from finite fields to the universal model over $\mathbb{Z}[1/2]$. Third, jet-scheme criteria give rational singularities and the lower bound for the ideal threshold. Maximal-isotropic strata provide the opposite bound. For rationality necessity, an elementary integrality obstruction first gives the expected complete-intersection dimension; isotropic strata and their first jets then exclude the entire complementary range.

The proofs of the uniform statements do not depend on computations. The ancillary code provides exact finite controls for the arithmetic, Fourier normalization, and first-jet counts.

2 Weighted counts of symmetric matrices

Let $\mathcal{O}$ be a discrete valuation ring with finite residue field of odd cardinality q , uniformizer π , and $R_\ell = \mathcal{O}/(\pi^\ell)$. Write $\nu_\ell(x)$ for the valuation of $x \in R_\ell$, with $\nu_\ell(0) = \ell$. Thus $|\pi^v R_\ell| = q^{\ell-v}$ for $0 \leq v \leq \ell$. For a square matrix A over R_ℓ , put

$$\rho_\ell(A) = \log_q |\operatorname{im} A|.$$

If its Smith factors have valuations a_i , allowing $a_i = \ell$, then $\rho_\ell(A) = \sum_i (\ell - a_i)$. Invertible row and column operations preserve ρ_ℓ . A matrix is called *primitive* if at least one entry is a unit; it need not be invertible.

2.1 Common valuation and scalar products

We first record the two counting facts used in the matrix estimates.

Lemma 2.1 (Removal of the common valuation). *Let $\mathcal{M}_k(R)$ denote either $\operatorname{Sym}_k(R)$ or $\operatorname{Hol}_k(R)$, the symmetric or hollow symmetric matrices, respectively. Every nonzero $A \in \mathcal{M}_k(R_L)$ determines a unique pair (ℓ, A_0) with*

$$A = \pi^{L-\ell} \tilde{A}_0, \quad 1 \leq \ell \leq L, \quad A_0 \in \mathcal{M}_k(R_\ell) \text{ primitive},$$

where the equality is independent of the lift $\tilde{A}_0$ to R_L . Here $L - \ell$ is the minimum valuation of the entries of A , and

$$\rho_L(A) = \rho_\ell(A_0). \tag{2.1}$$

Consequently, for every $\sigma > 0$,

$$\sum_{A \in \mathcal{M}_k(R_L)} q^{-\sigma \rho_L(A)} = 1 + \sum_{\ell=1}^L \sum_{\substack{A_0 \in \mathcal{M}_k(R_\ell) \\ A_0 \text{ primitive}}} q^{-\sigma \rho_\ell(A_0)}. \tag{2.2}$$

Proof. Multiplication by $\pi^{L-\ell}$ induces an additive bijection

$$\iota_\ell : R_\ell \longrightarrow \pi^{L-\ell} R_L.$$

Its injectivity follows because $\pi^{L-\ell} x = 0$ modulo π^L exactly when $x = 0$ modulo π^ℓ . Applied entrywise, this bijection identifies primitive matrices over R_ℓ with matrices of common valuation $L - \ell$ over R_L . Symmetry and zero diagonal entries are preserved. For a vector $x \in R_L^k$,

$$Ax = \iota_\ell(A_0 \bar{x}), \quad \bar{x} = x \bmod \pi^\ell.$$

Reduction onto R_ℓ^k is surjective, so $\operatorname{im} A = \iota_\ell(\operatorname{im} A_0)$ and (2.1) follows. The valuation strata are disjoint and exhaust the nonzero matrices, proving (2.2). There is no additional lift factor: different lifts of A_0 give the same matrix A after multiplication by $\pi^{L-\ell}$. $\square$

Lemma 2.2 (A scalar product count). *For $t \in R_\ell$, let $N_\ell(t) = |\{(b, c) \in R_\ell^2 : bc = t\}|$. Then*

$$N_\ell(t) = \begin{cases} (w+1)(q-1)q^{\ell-1}, & \nu_\ell(t) = w < \ell, \\ q^\ell + \ell(q-1)q^{\ell-1}, & t = 0. \end{cases} \quad (2.3)$$

In particular, $N_\ell(t) \leq (\ell+1)q^\ell$ for every t .

Proof. There are $(q-1)q^{\ell-v-1}$ elements b of valuation $v < \ell$. For each, multiplication by b has image $\pi^v R_\ell$ and a kernel of cardinality q^v . Thus $bc = t$ has exactly q^v solutions in c if $t \in \pi^v R_\ell$, and none otherwise. Each eligible valuation v contributes $(q-1)q^{\ell-1}$ pairs. For nonzero t the eligible valuations are $0, \dots, w$. For $t = 0$ they are $0, \dots, \ell-1$, and $b = 0$ contributes the additional q^ℓ pairs. $\square$

2.2 The symmetric-matrix recurrence

For a real number $\sigma > 0$, define

$$Z_k(\ell, \sigma) = \sum_{A \in \text{Sym}_k(R_\ell)} q^{-\sigma \rho_\ell(A)}, \quad a_k(\sigma) = \max_{0 \leq j \leq k} \left\{ \frac{j(2k-j+1)}{2} - \sigma j \right\}.$$

The zero-by-zero convention is $Z_0 = 1$ and $a_0 = 0$.

Lemma 2.3. *Set $C_0 = 1$ and $C_k = 1 + \binom{k+1}{2} C_{k-1}$ for $k \geq 1$. Then*

$$Z_k(\ell, \sigma) \leq C_k(\ell+1)^k q^{\ell a_k(\sigma)} \quad (2.4)$$

for every odd q , every $\ell \geq 1$, and every real $\sigma > 0$.

Proof. Let $P_k(\ell, \sigma)$ be the same sum restricted to primitive matrices. A primitive symmetric matrix has $v^t A v$ a unit for at least one of the vectors

$$e_i \quad (1 \leq i \leq k), \quad e_i + e_j \quad (i < j).$$

Indeed, if every diagonal entry is a nonunit, a unit off-diagonal entry gives such a vector because 2 is invertible. Each vector extends to a fixed unimodular basis, whose congruence action is a bijection preserving image length. These $\binom{k+1}{2}$ charts cover the primitive matrices. They may overlap; summing their nonnegative masses gives an upper bound.

In one chart write

$$A = \begin{pmatrix} a & b^t \\ b & C \end{pmatrix}, \quad a \in R_\ell^\times, \quad S = C - bb^t/a.$$

Invertible block elimination changes A into $\text{diag}(a, S)$, so $\rho_\ell(A) = \ell + \rho_\ell(S)$. Conversely, each $(a, b, S) \in R_\ell^\times \times R_\ell^{k-1} \times \text{Sym}_{k-1}(R_\ell)$ gives exactly one matrix in the chart, by $C = S + bb^t/a$. In particular S is unrestricted, including $S = 0$ and matrices all of whose entries are nonunits. Since the number of pairs (a, b) is $(q^\ell - q^{\ell-1})q^{(k-1)\ell} \leq q^{k\ell}$,

$$P_k(\ell, \sigma) \leq \binom{k+1}{2} q^{(k-\sigma)\ell} Z_{k-1}(\ell, \sigma). \quad (2.5)$$

Lemma 2.1 now gives the explicit recurrence

$$Z_k(L, \sigma) = 1 + \sum_{\ell=1}^L P_k(\ell, \sigma) \leq 1 + \binom{k+1}{2} \sum_{\ell=1}^L q^{(k-\sigma)\ell} Z_{k-1}(\ell, \sigma). \quad (2.6)$$

Induct on k . Separating $j = 0$ and replacing j by $j + 1$ in the other terms of the maximum gives

$$a_k(\sigma) = \max\{0, k - \sigma + a_{k-1}(\sigma)\}.$$

For every $1 \leq \ell \leq L$, including when $k - \sigma + a_{k-1}(\sigma) < 0$,

$$q^{\ell(k-\sigma+a_{k-1}(\sigma))} \leq q^{La_k(\sigma)}.$$

Use this inequality and $\sum_{\ell=1}^L (\ell + 1)^{k-1} \leq (L + 1)^k$ in (2.6). Since $a_k(\sigma) \geq 0$, the initial term 1 is absorbed by the stated value of C_k . $\square$

2.3 An exact hollow pivot count

Let $U_h(\ell, \sigma)$ be the weighted mass of primitive matrices in $\text{Hol}_h(R_\ell)$, for $h \geq 2$.

Lemma 2.4. *With $B_h = \binom{h}{2} C_{h-2}$, one has*

$$U_h(\ell, \sigma) \leq B_h(\ell + 1)^{2h-4} q^{\ell b_h(\sigma)}, \quad b_h(\sigma) = \max_{2 \leq r \leq h} \{\delta_h(r) - \sigma r\}. \quad (2.7)$$

Moreover,

$$\sum_{H \in \text{Hol}_h(R_L)} q^{-\sigma \rho_L(H)} = 1 + \sum_{\ell=1}^L U_h(\ell, \sigma). \quad (2.8)$$

Proof. Every primitive hollow matrix has a unit off-diagonal entry. Let $W_h(\ell, \sigma)$ be the weighted mass in the chart $H_{12} \in R_\ell^\times$. Permuting indices identifies the other $\binom{h}{2}$ charts with this one. Coverage and nonnegativity give

$$U_h(\ell, \sigma) \leq \binom{h}{2} W_h(\ell, \sigma). \quad (2.9)$$

No disjointness of these charts is required.

Put $k = h - 2$. In the fixed chart write

$$H = \begin{pmatrix} A & B \\ B^t & C \end{pmatrix}, \quad A = \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}, \quad a \in R_\ell^\times, \quad S = C - B^t A^{-1} B.$$

The invertible matrix

$$E = \begin{pmatrix} I_2 & -A^{-1}B \\ 0 & I_k \end{pmatrix} \quad \text{satisfies} \quad E^t H E = \begin{pmatrix} A & 0 \\ 0 & S \end{pmatrix}.$$

As A is invertible,

$$\rho_\ell(H) = 2\ell + \rho_\ell(S). \quad (2.10)$$

This identity uses the full image length of S , irrespective of its rank modulo π .

Fix $a \in R_\ell^\times$ and an arbitrary $S \in \text{Sym}_k(R_\ell)$. If column j of B is $(b_j, c_j)^t$, the condition that $C = S + B^t A^{-1} B$ have zero diagonal is exactly

$$b_j c_j = -a S_{jj}/2 \quad (1 \leq j \leq k).$$

These equations use disjoint pairs of variables. Each choice of B satisfying them determines a unique hollow C ; its off-diagonal entries impose no further conditions. Thus the fiber over (a, S) has exactly $\prod_{j=1}^k N_\ell(-a S_{jj}/2)$ elements. Combining this count with (2.10) gives the exact chart identity

$$W_h(\ell, \sigma) = q^{-2\sigma\ell} \sum_{a \in R_\ell^\times} \sum_{S \in \text{Sym}_k(R_\ell)} q^{-\sigma\rho_\ell(S)} \prod_{j=1}^k N_\ell(-a S_{jj}/2). \quad (2.11)$$

In particular this sum includes every singular or nonprimitive residual matrix S . The case $S_{jj} = 0$ is covered by the second line of (2.3).

Use $N_\ell(t) \leq (\ell + 1)q^\ell$ and $|R_\ell^\times| \leq q^\ell$ in (2.11), then apply (2.9). This yields

$$U_h(\ell, \sigma) \leq \binom{h}{2} (\ell + 1)^k q^{(h-1-2\sigma)\ell} Z_k(\ell, \sigma). \quad (2.12)$$

Lemma 2.3 supplies a second factor $(\ell + 1)^k$. The exponent becomes

$$h - 1 - 2\sigma + a_{h-2}(\sigma) = \max_{0 \leq j \leq h-2} \{\delta_h(j+2) - \sigma(j+2)\} = b_h(\sigma),$$

proving (2.7). Finally, apply Lemma 2.1 to hollow matrices to obtain (2.8). $\square$

Remark 2.5 (Boundary case and mixed valuations). For $h = 2$ the residual matrix is empty, so the product in (2.11) equals one and

$$U_2(\ell, \sigma) = (q^\ell - q^{\ell-1})q^{-2\sigma\ell}.$$

Thus $B_2 = 1$ and the length exponent $2h - 4 = 0$ cause no exception. For a nontrivial valuation example, over $\mathbb{Z}/27$ take

$$H = 3 \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 3 \\ 1 & 3 & 0 \end{pmatrix}.$$

Its common valuation is one. Lemma 2.1 replaces it by the displayed primitive matrix H_0 over $\mathbb{Z}/9$, without changing image length. The pivot $a = 1$ leaves $S = (-6) = (3)$ over $\mathbb{Z}/9$. Therefore

$$\rho_3(H) = \rho_2(H_0) = 4 + \rho_2((3)) = 5.$$

The Smith valuations of H are $(1, 1, 2)$. Its reduction modulo 3 is zero, while the primitive reduction has rank two; neither residue rank alone records the residual contribution to image length.

For comparison, (2.11) gives the following complete count in the chart $H_{12} \in (\mathbb{Z}/9)^\times$ for $h = 3$. There are six choices of a ; the three rows account for all $6 \cdot 9^2 = 486$ matrices in this chart.

Residual S	Choices of S	$N_2(-aS/2)$	Matrices H	$\rho_2(H)$
0	1	21	126	4
3 or 6	2	12	144	5
A unit	6	6	216	6

The middle row records precisely the nonzero nonunit residuals.

3 Fourier inversion and a uniform fiber estimate

A primitive additive character $\psi : R_L \rightarrow \mathbb{C}^\times$ means one whose restriction to the socle $\pi^{L-1}R_L$ is nontrivial. The pairing $(a, b) \mapsto \psi(ab)$ is nondegenerate: for $a \neq 0$, the nonzero ideal aR_L contains the socle. Such characters exist for these finite local principal ideal rings. For $H \in \text{Hol}_h(R_L)$ set

$$G_L(H) = q^{-Lh} \sum_{x \in R_L^h} \psi(x^t H x / 2).$$

Lemma 3.1. *For every symmetric H , including singular H ,*

$$|G_L(H)|^2 = q^{-\rho_L(H)}.$$

Proof. In the square of the absolute value change variables from (x, y) to $u = x - y$ and $v = (x + y)/2$. This is a bijection, and the exponent becomes $u^t H v$. Summing over v gives q^{Lh} if $Hu = 0$ and zero otherwise. Thus $|G_L(H)|^2 = |\ker H|/q^{Lh}$. $\square$

For $c \in R_L^e$, character orthogonality and the independence of the M rows of V give

$$\frac{|\Phi_{M,h}^{-1}(c)(R_L)|}{q^{L(N-e)}} = \sum_{H \in \text{Hol}_h(R_L)} \psi \left(- \sum_{i < j} H_{ij} c_{ij} \right) G_L(H)^M. \quad (3.1)$$

The normalization uses one coordinate per unordered pair; the factor $1/2$ in $x^t H x / 2$ ensures the same convention.

Proposition 3.2. *There is a constant K_h depending only on h with the following properties.*

1. *If $M > F(h)$ and $\epsilon = \min_{2 \leq r \leq h} (Mr/2 - \delta_h(r))$, then*

$$\left| \frac{|\Phi_{M,h}^{-1}(c)(R_L)|}{q^{L(N-e)}} - 1 \right| \leq K_h q^{-\epsilon} \quad (3.2)$$

for every $L \geq 1$ and every target c .

2. *For arbitrary $M \geq 1$, put $b = b_h(M/2)$ and $\gamma = e - \max(0, b)$. Then*

$$|\Phi_{M,h}^{-1}(c)(R_L)| \leq K_h (L+1)^{2h-3} q^{L(N-\gamma)}. \quad (3.3)$$

Proof. The term $H = 0$ in (3.1) is one. Apply the triangle inequality, Lemma 3.1, and Lemma 2.4. In the strict case $b = -\epsilon$, with $\epsilon \geq 1/2$ since M, h, r are integers. Thus the nonzero mass is at most

$$B_h \sum_{\ell=1}^L (\ell+1)^{2h-4} q^{-\epsilon \ell} \leq B_h q^{-\epsilon} \sum_{\ell \geq 1} (\ell+1)^{2h-4} 3^{-(\ell-1)/2}.$$

The last series is finite and depends only on h . Without strictness, bound every exponential factor by $q^{L \max(0, b)}$ and sum the polynomial factors. $\square$

3.1 Quantitative densities over local fields

Let $\mathcal{O}$ now be the valuation ring of a nonarchimedean local field with residue cardinality q odd. Give each $\mathcal{O}^a$ its Haar probability measure. For a graph G with g edges, define the locally constant function on $\mathcal{O}^g$

$$f_{G,L}(c) = q^{-L(N-g)} \#\{V \in R_L^{M \times h} : \Phi_{M,G}(V) = c \bmod \pi^L\}.$$

It is the density of the pushforward after averaging on residue balls of radius q^{-L} .

Proposition 3.3. *Suppose $M > F(h)$, and put $\epsilon = \min_{2 \leq r \leq h} (Mr/2 - \delta_h(r))$ and $d_h = 2h - 4$. There is a constant A_h depending only on h such that, for every such $\mathcal{O}$ and every graph G , the pushforward of Haar probability under $\Phi_{M,G}$ has a continuous density f_G , and*

$$\|f_G - 1\|_\infty \leq A_h q^{-\epsilon}, \quad (3.4)$$

$$\|f_G - f_{G,L}\|_\infty \leq A_h (L+2)^{d_h} q^{-\epsilon(L+1)}, \quad L \geq 0, \quad (3.5)$$

where $f_{G,0} = 1$. In particular, if $c - c' \in \pi^L \mathcal{O}^g$, then

$$|f_G(c) - f_G(c')| \leq 2A_h (L+2)^{d_h} q^{-\epsilon(L+1)}.$$

These statements hold in both equal and mixed characteristic.

Proof. First take $G = K_h$ and write μ for the pushforward measure. Characters of the compact additive group $\mathcal{O}^e$ have finite conductor. Those of exact conductor $\ell \geq 1$ are indexed by primitive hollow matrices over R_ℓ , after choosing a generating additive character of R_ℓ . The Fourier coefficient of the pushforward at such a character is $G_\ell(H)^M$: the M rows are independent and Haar-uniform. Lemmas 3.1 and 2.4 therefore give

$$\sum_{\chi \text{ of conductor } \ell} |\hat{\mu}(\chi)| \leq U_h(\ell, M/2) \leq B_h (\ell+1)^{d_h} q^{-\epsilon \ell}.$$

Since $\epsilon \geq 1/2$, these bounds are summable. Fourier inversion on each finite quotient identifies its partial sum through conductor L with $f_{K_h,L}$. The series converges uniformly to a continuous function f . For each residue ball B , the integrals $\int_B f_{K_h,L}$ equal the pushforward measure of B once L is at least its level. Uniform convergence gives the same identity for f , so f is the density of the pushforward.

For the tail, write $\ell = L + 1 + j$ and use $(L + j + 2)^{d_h} \leq (L + 2)^{d_h} (j + 1)^{d_h}$. Then

$$\sum_{\ell > L} (\ell + 1)^{d_h} q^{-\epsilon \ell} \leq (L + 2)^{d_h} q^{-\epsilon(L+1)} \sum_{j \geq 0} (j + 1)^{d_h} 3^{-j/2}.$$

This proves (3.5); summing from $\ell = 1$ gives (3.4), after increasing A_h if needed. The finite-level function is constant on each residue ball, which gives the stated modulus. For a general graph, integrate f over the unused $e - g$ target coordinates. The same averaging sends $f_{K_h,L}$ to $f_{G,L}$ and cannot increase a supremum norm. It preserves continuity because the target is compact. This proves all claims, including the case of a graph without edges. $\square$

Remark 3.4. For $h = 2$ the exponent $\epsilon = M - 1$ in the modulus cannot be increased. Indeed, for $M \geq 2$ the map is the dot product of two vectors in $\mathcal{O}^M$. The first vector has minimum coordinate valuation j with probability $(1 - q^{-M})q^{-Mj}$. Conditional on this event, its dot product with a Haar-uniform second vector is uniform on $\pi^j \mathcal{O}$. Consequently, with $\nu(0) = \infty$,

$$f(c) = (1 - q^{-M}) \sum_{j=0}^{\nu(c)} q^{-(M-1)j}.$$

In particular, $f(0) - f(\pi^L)$ is a positive constant times $q^{-(M-1)(L+1)}$. For general h , no optimality of the modulus is asserted. Its continuity is consistent with the general FRS theory [1]; the displayed conductor-dependent bound and its validity at every odd residue characteristic follow directly from the hollow-matrix estimate.

4 All coefficient fibers

We now prove Theorem 1.3. Fix L , put $R = \mathbb{Z}[1/2]$ and $d = L(N - e)$, and first consider the complete-graph map

$$f : \mathcal{X} = \mathbb{A}_R^{LN} \longrightarrow \mathcal{Y} = \mathbb{A}_R^{Le}.$$

The argument first determines the geometric fibers at closed points, then transfers their properties to all points, proves flatness, and finally removes the unwanted edge coordinates.

Proof of Theorem 1.3. For a finite field $\mathbb{F}_q$ of odd characteristic, use $R_L = \mathbb{F}_q[t]/(t^L)$ in Proposition 3.2. Fix a coefficient target c defined over $\mathbb{F}_q$ and apply (3.2) over every finite extension. If $\mathcal{F} = f^{-1}(c)$, the resulting estimate is

$$\left| \frac{\#\mathcal{F}(\mathbb{F}_{q^a})}{q^{ad}} - 1 \right| \leq K_h q^{-a\epsilon}, \quad a \geq 1, \quad (4.1)$$

with $\epsilon > 0$ independent of a . Choose a finite extension defining all geometric irreducible components of the reduced support of $\mathcal{F}$, and let a tend to infinity through multiples of that degree. Apply [12, Theorem 1] to the projective closures of its positive-dimensional components. The boundaries, pairwise intersections, and components of smaller dimension contribute lower-order terms by [12, Lemma 1]; a zero-dimensional reduced scheme instead has an eventually constant point count. Thus, if $D = \dim \mathcal{F}$ and b_D is the number of geometric components of dimension D , then $\#\mathcal{F}(\mathbb{F}_{q^a})/q^{aD} \rightarrow b_D$. Equation (4.1) forces $D = d$ and $b_D = 1$: the normalized count would otherwise tend to zero, to infinity, or to a positive integer different from one.

The fiber is defined by Le equations in $\mathbb{A}^{LN}$. Every irreducible component has dimension at least d , by Krull's height theorem. It follows that there is just one component. At every point of the fiber, the defining ideal has height Le and Le generators in a regular local ring. These generators form a regular sequence there. The fiber is therefore a local complete intersection and is Cohen–Macaulay.

We check reducedness separately, since point counts cannot detect nilpotents. Taking $r = h$ in (1.2) gives $F(h) \geq h - 1$, so $M > F(h)$ implies $M \geq h$. Over the algebraic closure, complete the prescribed constant off-diagonal target to an invertible symmetric $h \times h$ matrix by choosing its diagonal. This is possible since the determinant contains the product of the diagonal variables with coefficient one. Realize this matrix as $V_0^t V_0$ with V_0 of full column rank.

For a symmetric matrix Q , choose W with $V_0^t W = Q/2$; this is possible because V_0^t is surjective. Then $V_0^t W + W^t V_0 = Q$, so the differential, including its off-diagonal part, is surjective. At order $a \geq 1$ the Gram coefficient is

$$V_0^t V_a + V_a^t V_0 + \sum_{b=1}^{a-1} V_b^t V_{a-b}.$$

The sum is already known when V_a is chosen. Thus the off-diagonal equations can be solved successively. At the resulting point, the coefficient Jacobian is block lower triangular, with the same surjective off-diagonal differential on each diagonal block. It has rank Le , so this is a smooth point of the fiber. The unique component is consequently generically reduced. The Cohen–Macaulay coordinate ring has no embedded associated primes; generic reducedness therefore makes it reduced. This proves geometric integrality for every target over a finite field.

The map f is affine and of finite presentation, and $\mathcal{Y}$ is Noetherian, quasi-compact, and Jacobson. Consider the subset

$$U = \{y \in \mathcal{Y} : \mathcal{X}_{\bar{y}} \text{ is reduced, has one irreducible component, and has dimension } d\}.$$

Here $\bar{y}$ denotes a geometric point over y . By [16, Tags 0579, 055B, 05F9], each condition is locally constructible. Since $\mathcal{Y}$ is affine, it is quasi-compact and quasi-separated; [16, Tag 054E] therefore makes U constructible.

These constructibility results do not assume flatness. Every closed point of $\mathcal{Y}$ has a finite residue field of odd characteristic, so the preceding argument puts every closed point in U .

For completeness, a nonempty constructible subset of a Noetherian Jacobson scheme contains a nonempty locally closed subset, hence a nonempty open subset of some closed subscheme. Closed points are dense in that closed subscheme and remain closed in the ambient scheme. Such a subset therefore contains a closed point. Apply this observation to the constructible complement $\mathcal{Y} \setminus U$: it must be empty. We obtain integral geometric fibers of dimension d at all points, including generic points and targets with transcendental coordinates. The height argument above applies to each such fiber and gives the complete-intersection assertion. All these conclusions concern geometric fibers and persist under extension of their residue fields.

We verify the local dimension hypothesis for miracle flatness, including points in positive characteristic. Let x lie over y , and let $\mathfrak{p}$ be their common image in $\text{Spec } R$. Put

$$\eta = \dim \mathcal{O}_{\text{Spec } R, \mathfrak{p}} \in \{0, 1\}, \quad b = \text{trdeg}_{\kappa(\mathfrak{p})} \kappa(y), \quad t = \text{trdeg}_{\kappa(y)} \kappa(x).$$

The dimension formulas for polynomial rings over $\mathbb{Q}$ when $\eta = 0$ and over the discrete valuation ring $R_{(\mathfrak{p})}$ when $\eta = 1$ give

$$\begin{aligned} \dim \mathcal{O}_{\mathcal{X}, x} &= LN + \eta - b - t, \\ \dim \mathcal{O}_{\mathcal{Y}, y} &= Le + \eta - b, \\ \dim \mathcal{O}_{\mathcal{X}_y, x} &= d - t. \end{aligned}$$

The last equality uses the integral d -dimensional fiber over $\kappa(y)$. Thus the first local dimension is the sum of the other two. The target local ring is regular and the source local ring is regular, hence Cohen–Macaulay. Miracle flatness [16, Tag 00R4] now gives flatness of f . Its fibers are nonempty, so f is faithfully flat. After any field base change, flatness pulls the coordinate regular sequence of a target point back to the displayed coefficient equations. Thus each fiber is a complete intersection defined by that sequence.

For a graph G , put $g = |E(G)|$ and $m = L(e - g)$. Compose f with the coordinate projection $\mathcal{Y} \rightarrow \mathbb{A}_R^{Lg}$. Both maps are faithfully flat, so their composition is faithfully flat. To identify a geometric fiber, take any algebraically closed field k of characteristic different from two and a target $c_G \in k^{Lg}$. The complete targets extending c_G form

$$\mathcal{B} = \text{Spec } A \simeq \mathbb{A}_k^m, \quad A = k[z_1, \dots, z_m].$$

The graph fiber $Y_G = \text{Spec } B$ is precisely $\mathcal{X} \times_{\mathcal{Y}} \mathcal{B}$. Consequently B is faithfully flat over A , and every geometric fiber of $Y_G \rightarrow \mathcal{B}$ is integral of dimension d , by the result for f .

Let $K = \text{Frac}(A)$. For each nonzero $a \in A$, tensoring the injection $A \xrightarrow{a} A$ with the flat A -module B shows that multiplication by a is injective on B . Localization therefore embeds B into $B \otimes_A K$. This generic-fiber ring is a nonzero domain, since it is geometrically integral; hence B itself is a domain. Moreover,

$$\dim B = \text{trdeg}_k \text{Frac}(B) = m + \text{trdeg}_K \text{Frac}(B \otimes_A K) = m + d = L(N - g).$$

Here faithful flatness also ensures that A embeds in B . The graph map over k is flat, so the coordinate regular sequence of c_G pulls back to its Lg displayed coefficient equations in $k[V_0, \dots, V_{L-1}]$. They form a regular sequence defining Y_G . This proves geometric integrality, the complete-intersection property, and the dimension for every graph fiber. The argument includes $g = 0$ and $g = e$, with a polynomial ring in zero variables interpreted as k . $\square$

Corollary 4.1. *If $M > F(h)$, every ordinary graph-Gram fiber in characteristic zero has rational singularities.*

Proof. Every jet scheme of the ordinary fiber over c is the coefficient fiber over $(c, 0, \dots, 0)$. Theorem 1.3 makes all these schemes integral. The ordinary fiber is an integral complete intersection. Mustařa's theorem [14, Theorem 0.1] therefore gives rational singularities. Since a local complete intersection is Gorenstein, these are equivalently canonical singularities [14, discussion following Theorem 0.1]. $\square$

5 The exact ideal threshold

Lemma 5.1 (The threshold of a homogeneous ideal). *Let $J \subset \mathbb{C}[x_1, \dots, x_n]$ be a nonzero proper homogeneous ideal. Then $\text{lct}_{\mathbb{A}^n}(J) = \text{lct}_0(J)$.*

Proof. Choose a proper log resolution $\pi : \tilde{Y} \rightarrow \mathbb{A}^n$ of J that is an isomorphism over $\mathbb{A}^n \setminus V(J)$, and write $J\mathcal{O}_{\tilde{Y}} = \mathcal{O}_{\tilde{Y}}(-\sum_i a_i E_i)$ and $K_{\tilde{Y}/\mathbb{A}^n} = \sum_i k_i E_i$. The resolution formula [15, Proposition 1.5] gives

$$\text{lct}_x(J) = \min_{i: x \in \pi(E_i), a_i > 0} \frac{k_i + 1}{a_i},$$

with the minimum of an empty set equal to infinity. Thus, for each real $c \geq 0$, the locus

$$B_c = \{x : \text{lct}_x(J) \leq c\} = \bigcup_{i: a_i > 0, (k_i + 1)/a_i \leq c} \pi(E_i)$$

is closed: it is a finite union of images of closed sets under the proper map π . Homogeneity means that every dilation $x \mapsto \lambda x$, $\lambda \in \mathbb{C}^*$, preserves J . The local threshold is invariant under these isomorphisms, so B_c is invariant under dilation. If B_c is nonempty, it contains a closed point x , and closedness applied to $\lambda \mapsto \lambda x$ on $\mathbb{A}^1$ implies $0 \in B_c$.

The global threshold $c_* = \min_{a_i > 0} (k_i + 1)/a_i$ is finite and attained on this resolution, so B_{c_*} is nonempty. It follows that $\text{lct}_0(J) \leq c_*$. The reverse inequality follows because the local minimum uses a subset of the divisors in the global minimum. This proves the assertion without any reducedness or complete-intersection hypothesis on J . $\square$

Proof of Theorem 1.2. Set $\gamma = e - \max(0, b_h(M/2))$ as in Proposition 3.2. For each fixed L , apply (3.3) to the zero target over every finite extension of a finite field of odd characteristic. Lang–Weil gives

$$\dim(X_{M,h})_{L-1} \leq L(N - \gamma). \quad (5.1)$$

Here the subscript is jet order. For each fixed L , take the zero-target coefficient scheme over $\mathbb{Z}[1/2]$. Its fiber dimensions lie between 0 and LN . Each dimension level set is locally constructible by [16, Tag 05F9], hence constructible by [16, Tag 054E] because the target is affine. The locus where the dimension exceeds $L(N - \gamma)$ is a finite union of such level sets, hence constructible. It contains no closed point, by the finite-field bound. The Jacobson argument used above makes it empty. In particular the bound holds over $\mathbb{Q}$ and, by invariance of dimension under field extension, over $\mathbb{C}$.

This argument is applied separately for every L ; it does not require an infinite intersection of constructible sets.

The jet formula for an arbitrary ideal in a smooth ambient space [15, Corollary 0.2] is

$$\text{lct}(I_{M,h}) = N - \sup_{L \geq 1} \frac{\dim(X_{M,h})_{L-1}}{L}. \quad (5.2)$$

No complete-intersection or reducedness hypothesis is imposed on $I_{M,h}$ in this formula. Thus (5.1) gives $\text{lct}(I_{M,h}) \geq \gamma$. The identity

$$e - \delta_h(r) = \binom{h-r+1}{2}$$

shows that γ is the right side of (1.4).

For the reverse inequality, the zeroth jet in (5.2) gives $\text{lct}(I_{M,h}) \leq \text{height } I_{M,h} \leq e$. We need only consider the case in which the inner minimum in (1.4) is less than e . Put $u = \lfloor M/2 \rfloor$, the maximal isotropic dimension. To see that $u \leq h-2$, observe that

$$e - \left(\binom{k+1}{2} + \frac{M(h-k)}{2} \right) = \delta_h(h-k) - \frac{M(h-k)}{2}.$$

An inner value below e therefore implies $M < F(h)$. For $2 \leq r \leq h$, one has $r + 2h/r \geq r + 2 \geq 4$, so $F(h) \leq 2h - 3$. It follows that $u \leq h - 2$. The quadratic expression in k is minimized at $k = u$ (also at $k = u - 1$ when $M = 2u$), and its value is

$$\tau = \begin{cases} uh - u(u-1)/2, & M = 2u, \\ ((2u+1)h - u^2)/2, & M = 2u + 1. \end{cases} \quad (5.3)$$

Indeed, the successive difference is $k + 1 - M/2$. It is negative for $k < u$ and positive for $k \geq u$ when M is odd; when M is even, it vanishes at $k = u - 1$, giving the two adjacent minimizers.

Let S be the locally closed stratum of matrices whose columns span a maximal totally isotropic u -plane. For even M choose either component of $\text{OGr}(u, M)$; for $u = 0$ take $S = \{0\}$. The open spanning-frame bundle over this orthogonal Grassmannian has dimension

$$\dim S = \begin{cases} uh + u(u-1)/2, & M = 2u, \\ uh + u(u+1)/2, & M = 2u + 1. \end{cases} \quad (5.4)$$

For even M the first line says $\dim X_{M,h} \geq N - \tau$. The zeroth jet in (5.2) then gives the required upper bound.

Suppose $M = 2u + 1$, and write $r = h - u$. A relation among the Jacobian rows at $V \in S$ is exactly a hollow symmetric H with $VH = 0$. If $K = \ker V$ has dimension r , the symmetric matrices with image in K form a space of dimension $r(r+1)/2$. For example, after choosing a matrix P whose columns are a basis of K , they are precisely PAP^t with A symmetric; this parametrization is injective. Requiring the h diagonal entries to vanish imposes at most h linear conditions. Consequently

$$\text{corank Jac } \Phi_{M,h}|_V \geq \frac{r(r+1)}{2} - h = \binom{r}{2} - u =: \delta, \quad \dim T_V X_{M,h} \geq N - e + \delta.$$

The assumption $\tau < e$ implies $\delta - r/2 = e - \tau > 0$. Since the first jet over V is its tangent space, the first jet over S has dimension at least

$$\dim S + N - e + \delta = N + u^2 = 2(N - \tau).$$

Taking $L = 2$ in (5.2) gives $\text{lct}(I_{M,h}) \leq \tau$. This completes the equality.

Finally, $I_{M,h}$ is a nonzero proper homogeneous ideal. Lemma 5.1 therefore identifies its global threshold with its threshold at the origin. $\square$

6 Sharp rationality and the log-canonical boundary

Lemma 6.1 (Consequences of integrality). *If $X_{M,h}$ is integral, then $M \geq h$ and $X_{M,h}$ is a complete intersection of dimension $N - e$. These conclusions apply, in particular, if $X_{M,h}$ has rational singularities.*

Proof. Put $d_i = \sum_{a=1}^M v_{ai}^2$ in the coordinate ring of $X_{M,h}$. The off-diagonal Gram entries vanish in this ring, so

$$\prod_{i=1}^h d_i = \det(V^t V).$$

If $M < h$, the right side is identically zero. However, each d_i is nonzero in the coordinate ring: evaluate at the frame whose i th column is the first coordinate vector and whose other columns are zero. Thus the ring is not a domain, proving $M \geq h$.

For $M \geq h$, the frame V_0 consisting of the first h coordinate vectors belongs to $X_{M,h}$. At V_0 the differential of the (i, j) th equation, for $i < j$, is $w_{ij} + w_{ji}$. These e linear forms are independent, so V_0 is smooth of dimension $N - e$. Integrality therefore forces $\dim X_{M,h} = N - e$. Its ideal therefore has height e and is generated by e elements in a polynomial ring, which makes it a complete intersection.

Finally, rational singularities imply normality. All irreducible components of this affine cone contain its vertex. A normal scheme has a local domain at that vertex, so it has only one component; normality also implies reducedness. Hence it is integral. $\square$

Proof of Theorem 1.1. Corollary 4.1 gives rational singularities of every geometric fiber when $M > F(h)$. Theorem 1.3 at $L = 1$ also gives flatness and reduced fibers, so the map is FRS, for the complete graph and for every subgraph. Conversely, the FRS property implies rational singularities of the zero fiber. It remains to prove that rationality of the zero fiber forces $M > F(h)$. Suppose that $X = X_{M,h}$ has rational singularities and $M \leq F(h)$. Lemma 6.1 gives $M \geq h$, integrality, and the complete-intersection dimension $n = N - e$.

Set $u = \lfloor M/2 \rfloor$. The inequality $F(h) \leq 2h - 3$ follows from $r + 2h/r \geq r + 2 \geq 4$ for $2 \leq r \leq h$. Thus $M \leq 2h - 3$, and taking the integer part of $M/2$ gives $u \leq h - 2$. The successive difference $k + 1 - M/2$ shows that the inner quadratic minimum in (1.4) is attained at $k = u$. Its value is the number τ in (5.3). Directly from $e - \delta_h(r) = \binom{h-r+1}{2}$,

$$\tau = e - \max_{2 \leq r \leq h} \{\delta_h(r) - Mr/2\} \leq e, \tag{6.1}$$

where the inequality is equivalent to $M \leq F(h)$. This is an arithmetic identity; it does not require a primeness or normality threshold.

Let S be the maximal-isotropic spanning-frame stratum from (5.4), choosing either component when M is even. Its closure lies in the closed subset

$$Z = \{V \in X : V^t V = 0\}.$$

This is a proper subset, since the frame V_0 in Lemma 6.1 has $V_0^t V_0 = I_h$.

If $M = 2u$, equations (5.3) and (5.4) give

$$\dim S = uh + \frac{u(u-1)}{2} = N - \tau \geq N - e = n.$$

This contradicts $S \subset Z \subsetneq X$, since a proper closed subset of an integral n -dimensional variety has dimension less than n .

Suppose instead that $M = 2u + 1$, and put $r = h - u$. The Jacobian-relation calculation used in the proof of Theorem 1.2 applies at every $V \in S$: symmetric matrices with image in $\ker V$ form a space of dimension $r(r+1)/2$, and imposing zero diagonal adds at most h linear conditions. Writing $\delta = \binom{r}{2} - u$, we obtain $\dim T_V X \geq n + \delta$. Consequently,

$$\begin{aligned} \dim((X)_1|_S) &\geq \dim S + n + \delta \\ &= N + u^2 = 2(N - \tau) \geq 2n. \end{aligned}$$

On the other hand, Mustař's criterion [14, Theorem 0.1] makes $(X)_1$ irreducible. Its open part over the smooth locus of X has dimension $2n$, hence so does $(X)_1$. For the jet projection $p : (X)_1 \rightarrow X$, the subset $p^{-1}(Z)$ is closed and proper, and contains $(X)_1|_S$. The displayed dimension bound is again a contradiction. Both parity arguments include equality $M = F(h)$. Therefore $M > F(h)$, as required. The complete graph gives necessity for a condition uniform over all graphs and targets. $\square$

6.1 Comparison with the normal range

The rationality proof above is independent of the prime-threshold theorem. We now use [4, Theorems 1–2] to identify the normal complete-intersection range $M \geq D_{\text{prime}}(h)$ and compare it with the rationality and log-canonicity ranges.

We include the integer comparison to specify exactly where normality and rationality differ. Put $T_k = k(k-1)/2$.

Lemma 6.2. *For all $h \geq 2$, $D_{\text{rat}}(h)$ is either $D_{\text{prime}}(h)$ or $D_{\text{prime}}(h) + 1$. The latter occurs exactly when*

$$T_k + 2 \leq h \leq \lfloor k^2/2 \rfloor \quad \text{for some } k \geq 4. \quad (6.2)$$

Proof. The case $h = 2$ is immediate. For $k \geq 2$, comparison with consecutive odd squares gives

$$\begin{aligned} \left\lfloor \frac{2h+1-\sqrt{8h+1}}{2} \right\rfloor = h-k &\iff T_k < h \leq T_{k+1}, \\ \left\lfloor \frac{2h+1-\sqrt{8h-7}}{2} \right\rfloor = h-k &\iff T_k + 1 < h \leq T_{k+1} + 1. \end{aligned}$$

Indeed, the respective square comparisons are $(2k-1)^2 < 8h+1 \leq (2k+1)^2$ and $(2k-1)^2 < 8h-7 \leq (2k+1)^2$. On $T_k + 2 \leq h \leq T_{k+1}$ both floors are $h-k$; at $h = T_{k+1} + 1$ they are $h-k-1$ and $h-k$, respectively. Taking the minimum in (1.5) yields

$$\begin{aligned} D_{\text{prime}}(h) &= 2h - 2k + 1, & T_k + 2 \leq h \leq T_{k+1}, \\ D_{\text{prime}}(h) &= 2h - 2k, & h = T_{k+1} + 1. \end{aligned}$$

These intervals partition all $h \geq 3$. The successive difference of the expressions maximized in (1.2) is $-1 + 2h/(r(r+1))$. On the first interval we may take the maximizing index $r = k$, and obtain

$$F(h) - D_{\text{prime}}(h) = \frac{k^2 - 2h}{k} \in [-1, 1).$$

Thus $\lfloor F(h) \rfloor + 1$ exceeds $D_{\text{prime}}(h)$ precisely when $2h \leq k^2$. On the one-point interval the maximizer can be taken as $k + 1$, and $F(h) - D_{\text{prime}}(h) = -2/(k + 1) \in (-1, 0)$. The exceptional interval is nonempty precisely for $k \geq 4$. $\square$

Corollary 6.3. *In the normal range $M \geq D_{\text{prime}}(h)$,*

$$X_{M,h} \text{ is log canonical} \iff M \geq F(h).$$

The normal log-canonical but nonrational cases are precisely

$$h = 2a^2, \quad M = (2a - 1)^2, \quad a \geq 2.$$

Proof. Put $X = X_{M,h}$, a normal complete intersection of dimension $N - e$. By [8, Theorem 1.3], X is log canonical if and only if $(X)_{L-1}$ is equidimensional for every $L \geq 1$. Its Le coefficient equations in $\mathbb{A}^{LN}$ make every component have dimension at least $L(N - e)$, while the principal component, the closure of the jets over the smooth locus, has exactly that dimension. Thus equidimensionality is equivalent to $\dim(X)_{L-1} = L(N - e)$. By (5.2), these equalities for all L are equivalent to $\text{lct}(I_{M,h}) = e$. Theorem 1.2 makes this equivalent to $M \geq F(h)$. For log canonicity without rationality, Lemma 6.2 forces an exceptional prime size and $F(h) = M$. In its notation this means $k^2 = 2h$. Hence $k = 2a$, and the formulas follow. The exceptional interval excludes $a = 1$. $\square$

7 Jet dimensions and detection by the first jet

Corollary 7.1. *Put $c = \text{lct}(I_{M,h})$. For every $L \geq 1$ such that M or L is even,*

$$\dim(X_{M,h})_{L-1} = L(N - c).$$

For all M, h ,

$$c = N - \max \left\{ \dim X_{M,h}, \frac{\dim(X_{M,h})_1}{2} \right\}.$$

Proof. The upper bound at every length is (5.1). If $c = e$, Krull's height theorem gives equality for every L . Suppose $c < e$ and put $u = \lfloor M/2 \rfloor$. For $M = 2u$, the smooth isotropic stratum S in (5.4) has dimension $N - c$. Its length- L jets have dimension $L(N - c)$ and give the lower bound.

For $M = 2u + 1$ and $L = 2a$, take an algebraic chart of the smooth space of independent ordered isotropic u -frames A . It has dimension $(3u^2 + u)/2$. The rank-one quotient $A^\perp/A$ admits, locally, a representative z whose norm is an invertible regular function; it need not be normalized to 1. Put $r = h - u$. For arbitrary length- L jets $A(t)$ in the chart, arbitrary $u \times r$ matrices $Y(t)$ modulo t^L , and arbitrary rows $w(t)$ of length r modulo t^a , set

$$V(t) = [A(t) \mid A(t)Y(t) + t^a z(t)w(t)] \pmod{t^{2a}}.$$

All Gram equations vanish: the first u columns are isotropic and orthogonal to the other columns, and the remaining inner products are divisible by t^{2a} . The parameters are recovered from $V(t)$ by the first u columns and the independent coordinates $A(t), z(t)$. Thus this is a locally closed family of dimension

$$L \left(\frac{3u^2 + u}{2} + ur \right) + ar = L \left(uh + \frac{u(u + 1)}{2} + \frac{r}{2} \right) = L(N - c).$$

For $u = 0$ the same construction is simply $V(t) = t^a w(t)$. For an explicit chart, use the split form $\langle (x, y, \zeta), (x', y', \zeta') \rangle = x^t y' + y^t x' + \zeta \zeta'$. Writing $P \in \text{GL}_u$, $\Omega^t = -\Omega$, and C a $1 \times u$ row, take

$$A = \begin{pmatrix} P \\ (\Omega - C^t C/2)P \\ CP \end{pmatrix}, \quad z = \begin{pmatrix} 0 \\ -C^t \\ 1 \end{pmatrix}.$$

Direct multiplication verifies $A^t J A = 0$, $A^t J z = 0$, and $z^t J z = 1$, where J is the matrix of the split form. The parameters P, Ω, C give precisely the dimension of the isotropic-frame chart used above. This proves the even-length statement. The threshold formula using only orders zero and one follows by choosing $L = 1$ in the even- M case and $L = 2$ in the odd- M case, with the case $c = e$ already covered. $\square$

Corollary 7.2. *If $M \geq D_{\text{prime}}(h)$, then $X_{M,h}$ has rational singularities if and only if its first jet scheme is irreducible. It has log-canonical singularities if and only if its first jet scheme is equidimensional.*

Proof. Rationality and log canonicity give the stated jet properties by [14, 8]. If $M < D_{\text{rat}}(h)$ in the normal range, Lemma 6.2 gives an exceptional size $M = D_{\text{prime}}(h) = 2h - 2k + 1$. The isotropic jet family in the proof of Theorem 1.1 then has dimension at least $2 \dim X_{M,h} + (k^2 - 2h)$. At excess zero the first jet is reducible; at positive excess it is also not equidimensional, since its principal component still has dimension $2 \dim X_{M,h}$. These are all remaining cases by Lemma 6.2. $\square$

An odd-length example. The bound can be strict when both M and L are odd. For $M = 1$, $h = 4$, and $L = 3$, Theorem 1.2 gives $c = 2$, so (5.1) bounds $\dim(X_{1,4})_2$ by $3(4 - 2) = 6$. In fact its dimension is 5. Write its four coordinates as $x_i(t) = a_i + b_i t + c_i t^2$ modulo t^3 . If some $a_i \neq 0$, then $x_i(t)$ is a unit, and the equations $x_i(t)x_j(t) = 0$ force $x_j(t) = 0$ for every $j \neq i$. These four open strata have dimension 3. On the closed locus $a_1 = \dots = a_4 = 0$, the only remaining conditions are $b_i b_j = 0$ for $i < j$. The b -coordinates form a union of four coordinate axes, while the four c_i are free. This closed locus has dimension $1 + 4 = 5$, proving the claim.

8 Reproducible checks and further questions

The proof rests on Lemmas 2.3–3.1, Fourier inversion, and the cited geometric theorems. The four Python programs in the ancillary `anc/` directory supply bounded checks that can expose normalization or arithmetic errors:

- Exact threshold arithmetic was checked for $2 \leq h \leq 20,000$, with an independent brute-force maximization for $h \leq 600$. The minimization and isotropic-stratum arithmetic in (1.4) were checked for 23,095 pairs (h, M) . The direct parity obstruction in the rationality proof was compared with $M \leq F(h)$ for 179,101 pairs with $2 \leq h \leq 600$, without using the prime-threshold formula in that comparison.
- Exhaustive small-ring matrix enumeration includes all 531,441 hollow 4×4 matrices over $\mathbb{Z}/9$, 355,384 Schur-pivot checks, and selected independent kernel counts verifying image lengths.
- Direct vector enumeration and bit-set intersections give the same counts as exact Fourier inversion for 22 prescribed fibers in six cases, including both $\mathbb{Z}/9$ and $\mathbb{F}_3[t]/(t^2)$. Zero, nonzero, and nonunit targets are included.

- Four additional checks in odd ambient dimensions compare first-jet counts obtained from literal Jacobian ranks with signed Fourier inversion. They include $M = 3, h = 4$, whose ideal threshold is $11/2$.
- The standalone program `check_valuation_strata.py` computes image lengths by enumerating kernels, without Smith elimination. It checks all 3,048 hollow matrices in twelve small cases over $\mathbb{Z}/3^\ell$ and $\mathbb{F}_3[t]/(t^\ell)$, and compares the counts of 280 pivot fibers with (2.11). It also checks the exact common-valuation decomposition, (2.3), and the mixed-valuation example in Remark 2.5.

The review PDF carries all four scripts, their JSON results, the ancillary README and checksums as document attachments. It also embeds `gram_verification_code.zip`: save this file from the PDF reader's Attachments panel and extract it to obtain `anc/`. The accompanying review archive `sharp_gram_singularity_thresholds_review.zip` contains the same PDF, `main.tex`, and the complete `anc/` directory. The arXiv source archive supplies `main.tex` and `anc/`; its ordinary L^AT_EX build produces the typeset pages without the review PDF's document attachments. After extracting the code ZIP or either complete archive, run these commands from its root with Python 3.11 or later:

```
python -X utf8 anc/check_gram_fourier_arithmetic.py
python -X utf8 anc/check_direct_gram_counts.py
python -X utf8 anc/check_odd_gram_first_jets.py
python -X utf8 anc/check_valuation_strata.py
```

No third-party Python packages are required. Each script writes a JSON result beside itself, with **status:** `PASS` on successful completion and a SHA-256 hash of its producing script. Keep the four scripts together: the first-jet checker imports the ring implementation from the direct-count checker. The supplied JSON files record the finite cases and exact outputs; `anc/SHA256SUMS.txt` identifies the distributed scripts and results. Reruns change elapsed times, so numerical payloads should be compared after excluding the `seconds` field. The ancillary README gives the scope of each check. These computations supplement the symbolic proofs; they do not establish the geometric transfer or the cone lemma.

The positive-characteristic question can be stated more precisely using the prior results. Over an algebraically closed field of odd characteristic, F -rationality of the zero-fiber ring requires $M \geq D_{\text{rat}}(h)$. Indeed, F -rationality implies normality. All components of this homogeneous scheme meet the origin, so normality forces integrality and hence $M \geq D_{\text{prime}}(h)$ by [4, Theorem 2]. If $D_{\text{rat}}(h) = D_{\text{prime}}(h) + 1$, Lemma 6.2 places h in exactly the exceptional family for which [19, Theorem 1.3] excludes F -rationality at $M = D_{\text{prime}}(h)$ in every odd characteristic. On the other hand, [17, Theorem 3.2] supplies strong F -regularity when $M \geq 3h - 4$. Thus the remaining question is whether $M \geq D_{\text{rat}}(h)$ suffices, uniformly in odd characteristic, even for the zero fiber. Theorem 1.3 gives integral coefficient fibers in that range, but the point counts and the local-field density bounds do not establish the required Frobenius criteria.

A second question is a uniform formula for jet dimensions at odd lengths in odd ambient dimension outside the log-canonical range.

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